Assume/Guarantee Contract of `GyroStateSet`

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1 Variable Declaration

Maps : `variables` in Code $\mapsto$ variables in Formal Specification

Alphabet : powerState, healthState, StateFlag′, clock, clock′

Inputs :

- `powerState` $\mapsto$ powerState : $\mathbb{N}$
- `healthState` $\mapsto$ healthState : $\mathbb{N}$

Outputs :

- `StateFlag` $\mapsto$ StateFlag : $\mathbb{N} \rightharpoonup \mathbb{N}$

In-Outputs : None

States : None

Parameters : None

Clocks : clock, clock′ : $\mathbb{R}_{\geq 0}$

2 AG Contract

Assumptions : **True**

Guarantees :

$$Guarantees \triangleq Check_1 \wedge Check_2$$

$$Check_1 \triangleq \text{clock}' = \text{clock}$$

$$Check_2 \triangleq \forall 0 \leq i < 9 \cdot \begin{pmatrix} \text{if powerState}\%2^{i+1} \geq 2^i \ \wedge \ \text{healthState}\%2^{i+1} < 2^i \ \text{then} \\ \quad \text{StateFlag}'[i] = \texttt{0xEB} \\ \text{else} \\ \quad \text{StateFlag}'[i] = \texttt{0x00} \\ \text{endif} \end{pmatrix}$$